

1(a)(i)	<i>independent variable</i> type of blood vessel / artery <u>and</u> vein <u>and</u> <i>dependent variable</i> mass required to break (the blood vessel) ;	1
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Question	Answer	Marks												
1(a)(ii)	any six from: 1 method to cut the blood vessel ; 2 method to measure blood vessel of, same / stated, length / AW ; 3 <i>idea of</i> discard samples that are damaged ; 4 <i>idea of</i> waiting (to see if the blood vessel breaks) before adding the next (10 g) mass ; 5 use, artery and vein / blood vessels, from the same source ; 6 add (10 g) masses, to the mass hanger, until the blood vessel breaks ; 7 <i>idea of</i> measuring mass of mass hanger / including mass of mass hanger ; 8 for each type of blood vessel, measure / note / record / AW, (total) mass when the blood vessel broke ; 9 use at least three measurements for each type of blood vessel <u>and</u> calculate a mean ; 10 safety comment with hazard <u>and</u> precaution ; <table border="1"> <thead> <tr> <th>hazard</th><th>risk</th><th>precaution</th></tr> </thead> <tbody> <tr> <td>knife / scalpel</td><td>injury</td><td>cut away from hand</td></tr> <tr> <td>blood vessels</td><td>biohazard / pathogens / allergy</td><td>gloves / mask / PPE / use disinfectant / wash hands</td></tr> <tr> <td>mass hanger falling</td><td>injury</td><td>suitable precaution to prevent injury</td></tr> </tbody> </table> 11 <i>idea of</i> repeat procedure with smaller mass intervals near to breaking mass ;	hazard	risk	precaution	knife / scalpel	injury	cut away from hand	blood vessels	biohazard / pathogens / allergy	gloves / mask / PPE / use disinfectant / wash hands	mass hanger falling	injury	suitable precaution to prevent injury	6
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1(b)(i)	<table border="1"> <thead> <tr> <th>mass added / g</th><th>length of vein / mm</th><th>percentage increase in length of ring of vein</th></tr> </thead> <tbody> <tr> <td>0</td><td>21</td><td>0</td></tr> <tr> <td>10</td><td>36</td><td>71</td></tr> <tr> <td>20</td><td>38</td><td>81</td></tr> <tr> <td>30</td><td>40</td><td>90</td></tr> <tr> <td>40</td><td>41</td><td>95</td></tr> <tr> <td>50</td><td>41</td><td>95</td></tr> </tbody> </table> three values correct ;; one or two correct max 1	mass added / g	length of vein / mm	percentage increase in length of ring of vein	0	21	0	10	36	71	20	38	81	30	40	90	40	41	95	50	41	95	2
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1(b)(ii)	1 to make a valid comparison (with other blood vessels / veins) ; 2 allows comparison as, initial / starting, length (of veins) is, not constant / AW ;	1																					
1(b)(iii)	1 correct orientation for axes <u>and</u> linear scale ; 2 axes labelled <u>and</u> units; 3 all points plotted correctly <u>and</u> line drawn accurately ;	3																					
1(b)(iv)	1 curve for artery starts at same point as the vein ; 2 curve for artery below curve for vein <u>and</u> does not intersect ;	2																					

Question	Answer	Marks
1(b)(v)	any two from: 1 starts at, same point / origin, as using percentage increase ; 2 (muscular) arteries have a thicker<u>er</u>, wall / tunica media / muscle layer (than the vein) or 3 (muscular) arteries have more (smooth) muscle (than the vein) ; 4 <i>idea of</i> (muscular artery) withstands more (blood) pressure so stretches less (than the vein) ;	2
1(b)(vi)	any two from: 1 use smaller mass intervals ; 2 extend the range of the masses added ; 3 use a, force meter / Newton meter / data logger / AW, (to apply force) ; 4 attach a, pointer / fiducial mark, to the ring of blood vessel (to improve length measurement / removing parallax) ; 5 use, same / stated, width of blood vessel ;	2